

Shreyas RAORANE



PROFILE

Robotics software engineer with a foundation in real-time C++ and systems software, working across motion planning, autonomy, and simulation. M.S.E. from Penn’s GRASP Lab.

CONTACT DETAILS

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LANGUAGES

English (fluent)
Hindi (fluent)
Marathi (native)

SKILLS

Programming: C++, Python, MATLAB, OOD, Design Patterns
Systems: Multi-threading, Real-time Systems, Memory Management, Event-driven Architecture, Linux
Robotics: ROS2, Sensor Integration (LIDAR, IMU, RGB-D), Localization, Multi-Agent Coordination
Motion Planning: RRT*, A*, MPC, DMPC, QP Optimization, Sampling-Based Planning, Collision Avoidance
Tools & Sim: Gazebo, PyBullet, AutoDRIVE, Foxglove, Docker, Git, CI/CD

WORK EXPERIENCE

TIME-CRITICAL SOFTWARE ENGINEERING INTERN.
OPEX Corporation (Moorestown, NJ). **2025.06–2025.08**
◇ Built a multi-threaded discrete-event simulation in C++ managing 1,000+ parallel events across 3 threads, with a scheduling engine matching real hardware timing at millisecond scale.
◇ Refactored the legacy codebase around a centralized SimulationHandler, letting the team validate against the simulator instead of scarce physical hardware.

SOFTWARE DEVELOPMENT ENGINEER.
Brillio Technologies (Bengaluru, India). **2022.06–2024.06**
◇ Delivered production software for Verizon’s ESAP Migration Project (10,000+ users); optimized 40+ PostgreSQL queries, reducing response time by 35%.

EDUCATION

M.S.E. ROBOTICS.
University of Pennsylvania. **2024–2026**
◇ GPA 3.80/4.0, Summa Cum Laude.
B.TECH.
SRM Institute of Science and Technology. **2018–2022**
◇ GPA 3.91/4.0, First Class with Distinction.

RESEARCH

MASTER’S THESIS: SCALABLE MULTI-ROBOT COORDINATION.
Penn GRASP Lab. **2025.10–2026.05**
◇ Advisors: Prof. Linh Phan, Prof. Rahul Mangharam, Prof. M. Ani Hsieh.
◇ Designed LCBA (Local Consensus Bundle Algorithm), a decentralized task allocation algorithm extending GCBBA to intermittently disconnected communication graphs; maintains non-zero task completion where CBBA and SGA baselines drop to zero.
◇ Integrated LCBA with three path planners (Cooperative A*, PBS, RHCR); evaluated across six maps and 6–200 agents over 432+ runs, reaching a 0.91 makespan ratio vs. centralized SGA.

ROBOTICS PROJECTS

F1/10 AUTONOMOUS RACING STACK.
University of Pennsylvania. **2025.02–2025.03**
◇ Built a full ROS2 autonomy stack integrating LIDAR, IMU, and RGB-D for perception, localization, and planning; bridged the AutoDRIVE Simulator and F1/10 hardware over a shared message interface with live parameter tuning.
SBAMP: SAMPLING-BASED ADAPTIVE MOTION PLANNING.
University of Pennsylvania. **2025.03–2025.04**
◇ Built an adaptive RRT* planner with dynamical-systems attractors for real-time goal tracking, replanning fast enough to track goals at speeds where static RRT* could not keep up.
SWARMSYNC: DISTRIBUTED MPC FOR QUADROTOR SWARMS.
University of Pennsylvania. **2025.10–2025.12**
◇ Implemented distributed MPC with QP optimization for collision-free trajectories across 8+ quadrotors, reaching goals reliably among dynamic obstacles.